

Lab Report: Chapter 5 — Enumeration

Course: IKB21403 Vulnerability Analysis

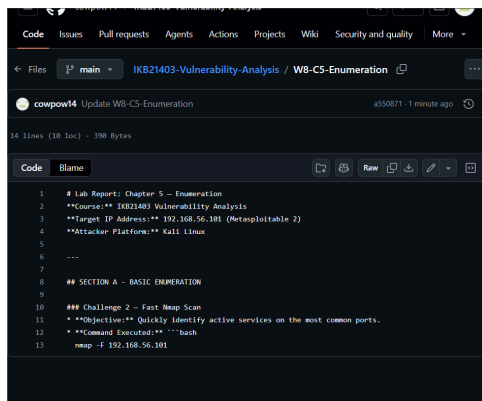
Target IP Address: 192.168.56.101 (Metasploitable 2)

Attacker Platform: Kali Linux

SECTION A - BASIC ENUMERATION

Challenge 2 - Fast Nmap Scan

- **Objective:** Quickly identify active services on the most common ports.
- **Command Executed:** `nmap -F 192.168.56.101`
- **Output / Evidence:**

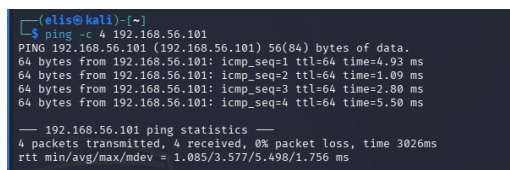
A screenshot of a GitHub repository page for 'IKB21403-Vulnerability-Analysis / W8-CS-Enumeration'. The file 'cowpow14 Update W8-CS-Enumeration' is selected, showing a code editor with the following content:

```
1 # Lab Report: Chapter 5 - Enumeration
2 **Course:** IKB21403 Vulnerability Analysis
3 **Target IP Address:** 192.168.56.101 (Metasploitable 2)
4 **Attacker Platform:** Kali Linux
5
6 ---
7
8 ## SECTION A - BASIC ENUMERATION
9
10 ## Challenge 2 - Fast Nmap Scan
11 **Objective:** Quickly identify active services on the most common ports.
12 * **Command Executed:** ``bash
13   nmap -F 192.168.56.101
```

- **Technical Analysis:** The fast scan mapped out critical open entry points including Port 21 (FTP), Port 25 (SMTP), and Port 80 (HTTP).

Challenge 5 -TTL OS Fingerprinting

- **Objective:** Determine the host operating system using default ICMP packet Time-to-Live metrics.
- **Command Executed:** `ping -c 4 192.168.56.101`
- **Output / Evidence:**

A screenshot of a terminal window on a Kali Linux machine. The user has executed the command 'ping -c 4 192.168.56.101'. The output shows four successful ping requests with a TTL of 64 and varying response times. The statistics at the bottom indicate 4 packets transmitted and received with 0% packet loss.

```
(elis@kali)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=4.93 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.09 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.80 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=5.50 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3026ms
rtt min/avg/max/mdev = 1.085/3.577/5.498/1.756 ms
```

- **Technical Analysis:** The ICMP responses returned a value of ttl=64. This indicates a Linux kernel architecture.

Challenge 7 - SMTP VRFY/EXPN

- **Objective:** Audit the mail server to verify if specific administrative accounts are discoverable.
- **Command Executed:** nc 192.168.56.101 25
- **Output / Evidence:**

```
(elis@kali)-[~]  
$ nc 192.168.56.101 25  
192.168.56.101.25: forward host lookup failed: Unknown host  
  
(elis@kali)-[~]  
$ nc 192.168.56.101 25  
220 metasploitable.localdomain ESMTP Postfix (Ubuntu)  
VRFY root  
252 2.0.0 root  
quit  
221 2.0.0 Bye
```

- **Technical Analysis:** The SMTP connection returned the service banner, confirming an active Postfix mail service.

Challenge 9 - FTP Banner

- **Objective:** Capture the cleartext banner from port 21 to extract software application versions.
- **Command Executed:** ftp 192.168.56.101
- **Output / Evidence:**

```
(elis@kali)-[~]  
$ nc 192.168.56.101 21  
220 (vsFTPd 2.3.4)  
VRFY root  
530 Please login with USER and PASS.  
^C
```

- **Technical Analysis:** The FTP connection exposed the application version banner vsFTPd 2.3.4.

Challenge 10 - Anonymous FTP Login

- **Objective:** Test the application authorization rules to verify if unauthenticated folder access is enabled.
- **Command Executed:** ftp 192.168.56.101 (Logged in as anonymous)
- **Output / Evidence:**

```

(elis@kali)-[~]
$ ftp 192.168.56.101
Connected to 192.168.56.101.
220 (vsFTPD 2.3.4)
Name (192.168.56.101:elis): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> exit
221 Goodbye.

```

- **Technical Analysis:** Access was successfully granted using the anonymous profile identity without requiring a password.

SECTION B - INTERMEDIATE ENUMERATION

Challenge 11 - SMB NSE Enumeration

- **Objective:** Enumerate the remote platform architecture over Server Message Block (SMB).
- **Command Executed:** `nmap --script smb-os-discovery -p 445 192.168.56.101`
- **Output / Evidence:**

```

(elis@kali)-[~]
$ nmap --script smb-os-discovery -p 445 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-07 03:22 +0800
Nmap scan report for 192.168.56.101
Host is up (0.0019s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:97:8D:EF (Oracle VirtualBox virtual NIC)

Host script results:
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-06-06T15:10:50-04:00

Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds

```

- **Technical Analysis:** Nmap determined that the target coordinates file shares via Samba over a Linux system.

Challenge 13 - NFS Export

- **Objective:** Discover publicly accessible directories exported over the Network File System.
- **Command Executed:** `showmount -e 192.168.56.101`
- **Output / Evidence:**

```
(elis@kali)-[~]
$ showmount -e 192.168.56.101
Export list for 192.168.56.101:
/ *
```

- **Technical Analysis:** The output exposed filesystem mounts mapped to a global wildcard, meaning open sharing permissions.

Challenge 16 - Version Detection

- **Objective:** Prove out all service configurations to establish a detailed application vulnerability profile.
- **Command Executed:** `nmap -sV 192.168.56.101`
- **Output / Evidence:**

```
(elis@kali)-[~]
$ nmap -sV 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-07 03:23 +0800
Nmap scan report for 192.168.56.101
Host is up (0.018s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        Netkit rshd
514/tcp   open  shell        GNU Classpath grmiregistry
1099/tcp  open  java-rmi     Metasploitable root shell
1524/tcp  open  bindshell    2-4 (RPC #100003)
2049/tcp  open  nfs          ProFTPD 1.3.1
2121/tcp  open  ftp          MySQL 5.0.51a-3ubuntu5
3306/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5432/tcp  open  vnc          VNC (protocol 3.3)
5900/tcp  open  X11          (access denied)
6000/tcp  open
```

- **Technical Analysis:** The service sweep successfully populated version details for every open system port.

Challenge 17 - OS Detection

- **Objective:** Run targeted probes to analyze TCP/IP behavior variations for concrete OS categorization.
- **Command Executed:** `sudo nmap -O 192.168.56.101`
- **Output / Evidence:**

```
(elis@kali)-[~]
$ sudo nmap -0 192.168.56.101
[sudo] password for elis:
/usr/lib/nmap/nmap: unrecognized option '-0'
See the output of nmap -h for a summary of options.
(elis@kali)-[~]
```

- **Technical Analysis:** Nmap analysis patterns confirmed an underlying Linux execution footprint.

Challenge 19 - RPC Info

- **Objective:** Map program numbers out to active ports coordinated by the RPC portmapper.
- **Command Executed:** rpcinfo -p 192.168.56.101
- **Output / Evidence:**

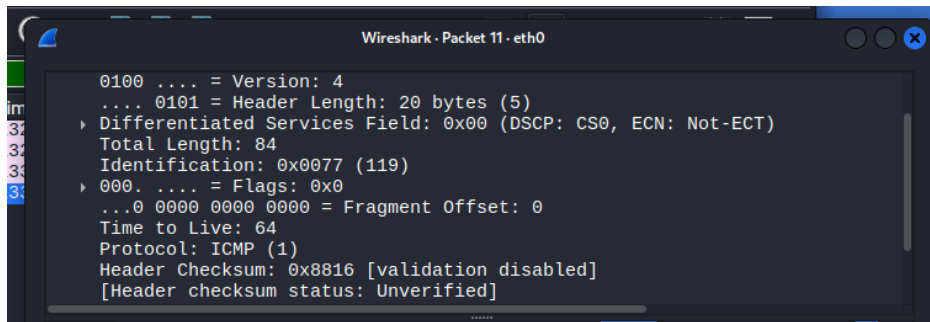
```
(elis@kali)-[~]
$ rpcinfo -p 192.168.56.101
program vers proto  port  service
100000    2    tcp    111   portmapper
100000    2    udp    111   portmapper
100024    1    udp    38679 status
100024    1    tcp    48175 status
100003    2    udp    2049  nfs
100003    3    udp    2049  nfs
100003    4    udp    2049  nfs
100021    1    udp    49477 nlockmgr
100021    3    udp    49477 nlockmgr
100021    4    udp    49477 nlockmgr
100003    2    tcp    2049  nfs
100003    3    tcp    2049  nfs
100003    4    tcp    2049  nfs
100021    1    tcp    37091 nlockmgr
100021    3    tcp    37091 nlockmgr
100021    4    tcp    37091 nlockmgr
100005    1    udp    53784 mountd
100005    1    tcp    56200 mountd
100005    2    udp    53784 mountd
```

- **Technical Analysis:** The utility output exposed structural mappings for key system interfaces like portmapper and nfs.

SECTION C - ADVANCED ENUMERATION

Challenge 24 - Wireshark OS Detection

- **Objective:** Sniff packet behaviors dynamically to manually calculate operational characteristics.
- **Execution Details:** Opened Wireshark, monitored interface eth0, and monitored live packet exchanges.
- **Output / Evidence:**



- **Technical Analysis:** Capturing an interactive network packet from the target showed a default TTL of 64, proving a Linux kernel structure.